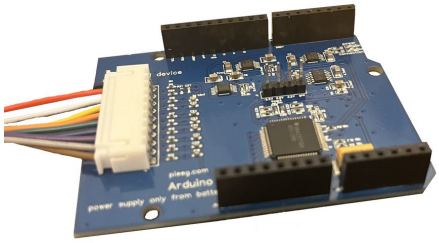
PCB & MANUFACTURING SPECIFICATION · SKU CQA240229AR

Board Material & Fabrication Data

Bare-board specification, mechanical features and the production quality test for ardEEG (board revision V1.0). Circuit design files are proprietary and are not part of this document.



FR-4	2	1.6 mm	68.6 × 53.3	0.25 mm
Base material	Copper layers	Board thickness	mm · UNO outline	Minimum track

01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCB, as used by common PCB order forms (Elecrow, JLCPCB, PCBWay).

Parameter	Specification	Note
Base material	**FR-4**, standard Tg (130–140 °C)	Glass-epoxy laminate
Layers	**2** (top and bottom copper)	
Dimensions	**68.6 × 53.3 mm**	Arduino UNO shield outline with chamfered corners
Thickness	**1.6 mm**	Standard; matches Arduino shield practice
Solder mask colour	**Blue**	As on production boards
Silkscreen	**White**	E1–E8, Ref, Biasout, pin labels, CE / UKCA, warnings
Surface finish	**ENIG** recommended	Flat pads for the 0.5 mm-pitch TQFP-64; lead-free HASL acceptable
Copper weight	**1 oz** (35 µm) outer layers	
Minimum track / spacing	0.25 mm (10 mil)	Standard capability
Minimum hole	Ø0.49 mm	Vias
RoHS	Lead-free materials and process	

02 LAYER STACK-UP

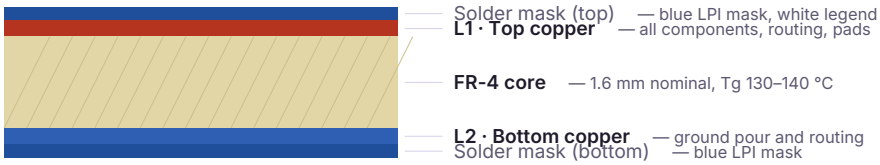


Figure 1 — 2-layer, 1.6 mm build. Layer roles are described in general terms only.

03 MECHANICAL FEATURES

Feature	Size	Qty	Purpose
Mounting holes	Ø3.18 mm	4	Arduino UNO hole pattern
Arduino header holes	Ø0.90 mm plated	32	Stacking headers J2, J4, J5, J6 on the UNO grid
Electrode header holes	Ø1.00 mm plated	10	1 × 10 electrode header, 2.54 mm
Test header holes	Ø1.00 mm plated	4	J1 supply test socket
Vias	Ø0.49 mm	85	Layer transitions

For research and education only. Not a medical device. Specifications subject to change.

04 ASSEMBLY

Technology	Single-sided SMT (all parts on the top side) plus through-hole headers
Key IC	TI ADS1299 (TQFP-64, 0.5 mm pitch) with low-noise on-board supplies
Passives	0402 and 0805 chip parts; one tantalum bulk capacitor
Connectors	4 long-leg stacking headers (10, 8, 8, 6 pins); 10-pin shrouded electrode header; 4-pin test socket
Indicators	2 green LEDs
Soldering	Lead-free reflow for SMD; selective or hand soldering for THT; no-clean flux, keep the analog input area free of residue
Marking	“ardEEG V1.0”, “power supply only from battery”, “not medical device”, CE, UKCA

05 PRODUCTION QUALITY TEST

1 Supplies

Mount on an Arduino UNO R4 WiFi powered from a battery. **LED D1** must light; J1 must read about −2.5 V, +2.5 V and 3.2 V against GND.

2 Data

Upload the standard sketch; **LED D2** lights and the Python receiver shows all 8 channels. The speed test must report 250 samples per second.

3 Noise

Set every CHnSET to 0x01 (input shorted). Each channel must stay within a few µV (≈ 0.5 µV RMS).

06 HANDLING, STORAGE AND COMPLIANCE

Topic	Requirement
ESD and storage	ESD-sensitive: handle by the edges on an anti-static surface. Store dry in the anti-static bag, 10–35 °C.
Environment	Indoor use only. No conformal coating; keep away from liquids and conductive gel.
Classification	Development and evaluation board for research and education. Not a medical device.
Design files	Schematic, layout, Gerbers and BOM are proprietary to PiEEG and are not published.

07 RELATED DOCUMENTS

AE-DS-001 Product Datasheet	**AE-MD-002** Mechanical & Dimensional Drawing	**AE-PIN-004** Connector Pinout & Electrode Guide
AE-QSG-005 Quick Start & Safety Guide	**AE-SW-006** Data Interface & Software Guide	